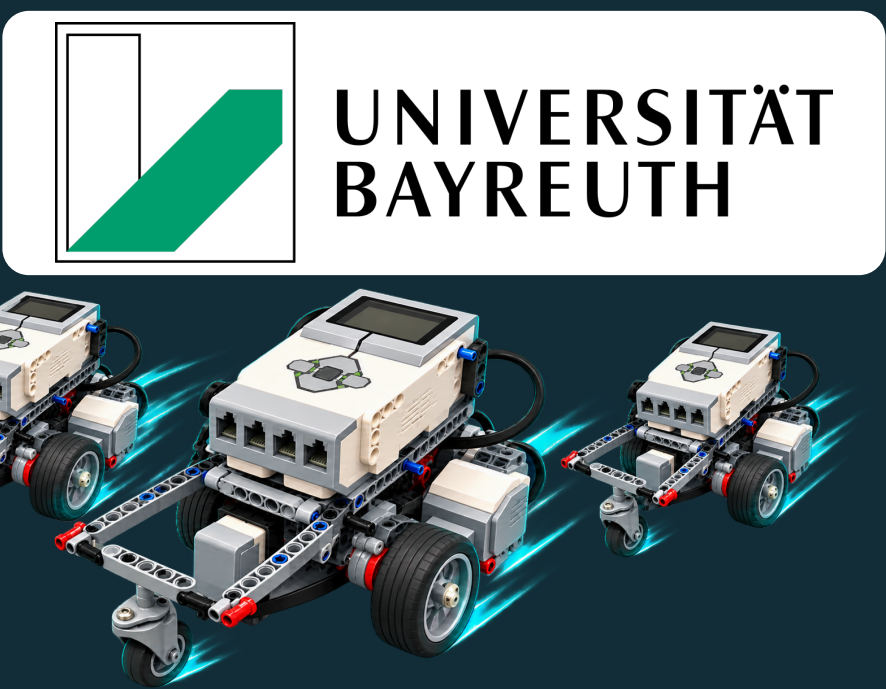


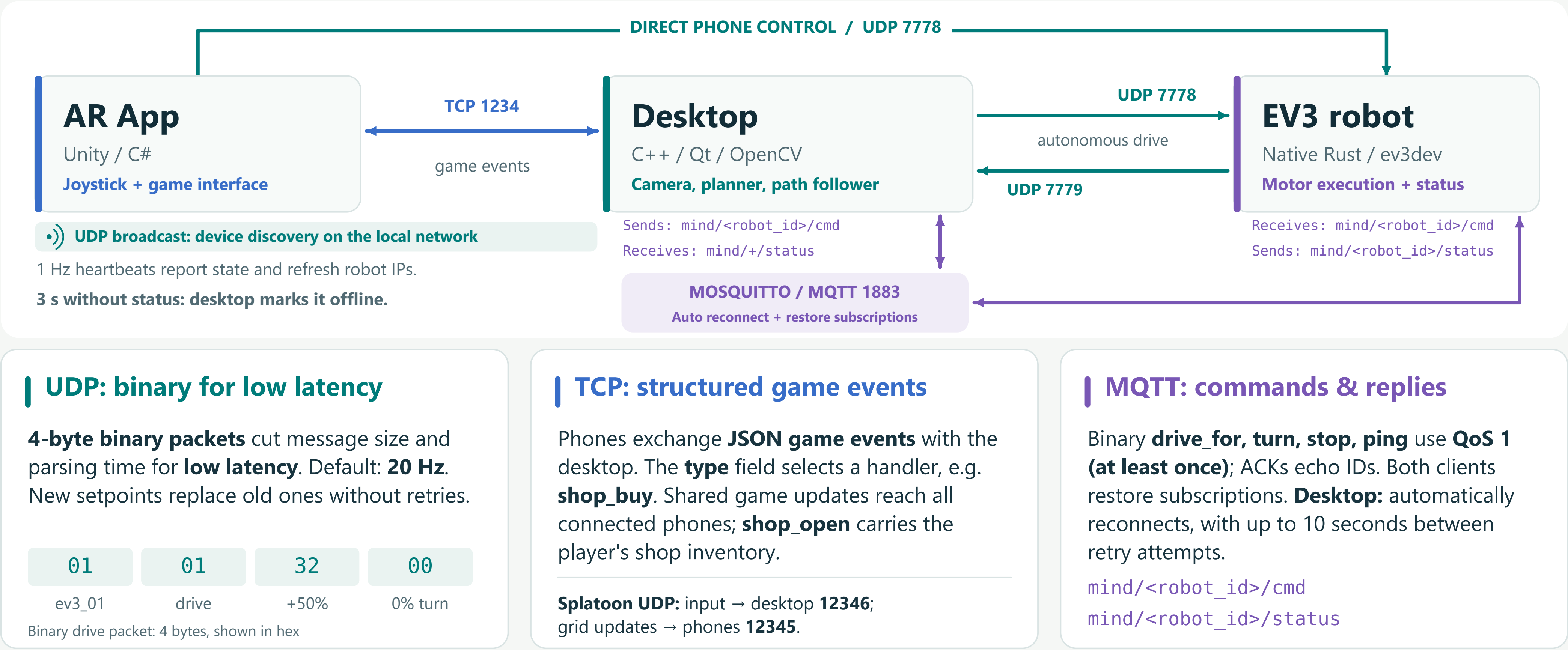


Robot Control & Communication

Collision avoidance, steering & network protocols

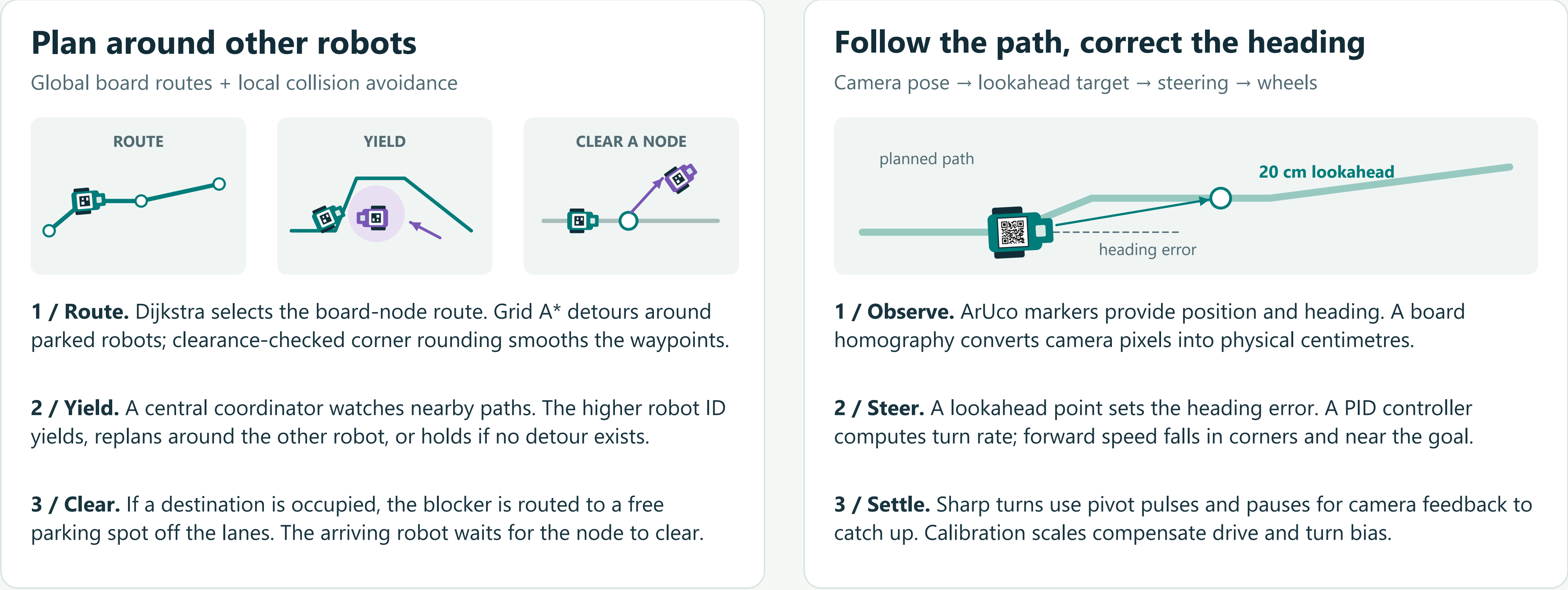
01 One system. Three communication channels.

ARCHITECTURE



02 From a target to controlled movement

DESKTOP CONTROL



03 Inside the brick

RUST + EV3DEV

Ferris at the wheel

Four workers, one motor controller

Native Rust runs directly on the EV3's Linux system. A mutex protects shared drive state; each valid UDP packet overwrites the setpoint and wakes the motor worker. A 10 ms timeout keeps watchdog checks running between packets.

UDP
Decode + ID filter
Wake the motor loop

MOTOR
10 ms check interval
Regulated B/C motors

MQTT
Dispatch commands
Interruptible moves

HEARTBEAT
1 Hz to desktop
State + battery + ID

Motion commands: stream speed + turn for continuous driving; queue **drive_for** and **turn** for distance and angle moves. The motor worker sends a completion ACK for queued moves.

Differential drive: $v_L = v - \omega b/2$; $v_R = v + \omega b/2$.
 v : forward speed; ω : turn rate; b : wheel spacing.
At the speed limit, both wheels slow together to keep the curve.
Motors B/C; 56 mm wheels; $b = 126$ mm.

Independent stop mechanisms

150 ms

No fresh drive packet: the EV3 watchdog stops continuous motion.

400 ms

Stale camera pose: the desktop commands zero drive.

STOP

MQTT sets an interrupt flag for precision moves; motors poll it.

